Sanskar Jain

C# / .NET Engineer | AI-Integrated Backend | SQL & NoSQL Data Systems
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Summary

C# and .NET backend engineer with 2+ years of industry experience building complex, production-grade applications, REST Web APIs, and event-driven microservices. Solid proficiency in SQL and NoSQL (MongoDB, PostgreSQL) with hands-on DB optimisation, indexing, and query tuning. Hands-on experience integrating LLM-based workflows, prompt engineering, and tool-calling patterns into backend services. Comfortable with Docker containerisation, unit testing, and end-to-end AI application design from backend API to lightweight frontend integration.

Skills

Languages: C#, Java, Python, JavaScript, TypeScript
Backend & AI: .NET 8, ASP.NET Core, REST Web APIs, Microservices, Prompt Engineering, LLM Tool Calling, RAG Pipelines
Databases: SQL (PostgreSQL, MySQL) — indexing, query optimisation, stored procedures; NoSQL (MongoDB)
Cloud & DevOps: Docker, Kubernetes, AWS (S3, SQS, SNS, Lambda), Azure DevOps, CI/CD, Terraform
Testing & Tools: Unit Testing, xUnit, Swagger, Postman, Splunk, Jira, Bitbucket

Experience

Verisk Analytics Jan 2024 – Present
Software Engineer II Hyderabad, India

- Developed and maintained complex **C# / .NET 8** ASP.NET Core applications, including production REST Web APIs handling **200K+ weekly requests**, with structured error handling, input validation, and Swagger-documented contracts.
- Achieved **34% improvement** in API response time through advanced **SQL query optimisation**, index tuning, connection pooling, and Redis/ElastiCache caching on hot-path endpoints across PostgreSQL and MongoDB data stores.
- Built **prompt-engineered LLM tool-calling workflows** and internal automation agents, integrating backend services with LLM APIs to automate Jira ticket creation and operational routing, reducing manual effort across teams.
- Engineered a scalable **MongoDB**-backed import/export pipeline for **30+ data types** using async processing and AWS S3 presigned URLs, cutting data transfer time by **38%** and reducing user-reported issues by **25%**.
- Migrated legacy services to **.NET 8**, modernising DI, async patterns, and SDK tooling — reducing technical debt by **55%** and improving runtime performance by **30%**; added xUnit test coverage as part of each migration.
- Built **RulesRunner**, a **C# SQS-backed microservice** that offloaded estimate compliance checks server-side, reducing client-side wait from **3000ms to 500ms** (83% reduction) and decoupling rules logic for flexible scaling.
- Containerised services using **Docker** and Podman with custom Dockerfiles; deployed to **Kubernetes (EKS/ECS)**, ensuring consistent environments across dev, staging, and production.
- Instrumented services with structured logging and **Splunk** dashboards, cutting mean time to diagnose production incidents and surfacing latency spikes and pipeline failures in real time.
- Slashed deployment time by **50%** by overhauling **Azure DevOps** CI/CD pipelines with staged gates, parallel jobs, and automated **unit test** integration, eliminating all manual release steps.

Projects

End-to-End AI Application with RAG Pipeline 2025 – 2026
Python, LLM APIs, Vector DB, REST API, Docker

- Designed and implemented a backend AI application integrating **RAG** with external data sources — embedding documents, querying a vector store, and surfacing context-aware responses via a REST API.
- Implemented **prompt engineering**, tool-calling chains, and external data integration patterns; packaged the service in a **Docker** container with a lightweight frontend interface.

Expense Tracker – Full-Stack Microservices Application Aug 2023 – Dec 2023
Spring Boot, Kafka, Docker, Kong, PostgreSQL

- Built 5+ microservices with **PostgreSQL** as the primary relational store, applying schema design, indexing, and **SQL query tuning** to handle 1000+ simulated transactions reliably.
- Containerised all services with **Docker**, reducing environment setup time by **60%**; integrated Kong API Gateway for routing, load balancing, and authentication.

Education

B.Tech in Computer Engineering — Vishwakarma Institute of Technology, Pune (SPPU) 2024 | CGPA: 8.8